

Social Domain Quantitative Data: Seven Empires Compared

Infonomics Project — CAS Framework Domain Completeness Retrofit

Domain 4: Social/Class Structure

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Executive Summary

This comprehensive data compilation reveals striking variation in social mobility mechanisms across pre-modern empires, with the Tang examination system achieving **1-2% pass rates** among approximately 2,000 annual candidates, the Ottoman devşirme system processing **200,000-300,000 boys** over three centuries, and Venice maintaining an essentially closed patriciate of **200-250 families** for 500 years. Gini coefficient estimates exist for only four of the seven target empires, ranging from **0.39-0.44 for Rome** to **0.70-0.76 for Venice**, with the Abbasid Caliphate and Mongol Empire lacking any scholarly inequality estimates due to insufficient source materials.

I. ROMAN EMPIRE (27 BCE – 476 CE)

A. Elite Composition and Senatorial Class Evolution

The Roman Senate maintained remarkable numerical stability under the Empire following Augustus's reforms. The body standardized at approximately **600 members** after Augustus reduced it from Julius Caesar's inflated 900. The senatorial order required a property qualification of **1,000,000 sesterces**, necessarily held in Italian land, establishing a wealth threshold roughly **1,111 times** a legionary's annual pay of 900 sesterces [HIGH confidence].

The equestrian order formed a substantially larger secondary elite, with estimates ranging to approximately **30,000 men** across the empire at its peak in the second century CE. Equestrian status required **400,000 sesterces** in property [HIGH confidence].

Period	Senate Size	Notes	Confidence
Republic (post-Kings)	300	Standard Republican size	HIGH
Sulla's reforms (80s BCE)	600	Doubled through equestrian inclusion	HIGH
Julius Caesar	900	Dramatic expansion	HIGH
Augustus onward	600	Stabilized	HIGH
Late Empire (4th century)	4,000+	Combined Eastern/Western	MODERATE

B. Citizenship Expansion

The Augustan censuses recorded **4,063,000 citizens** in 28 BCE, rising to **4,937,000** by 14 CE. The *Constitutio Antoniniana* of 212 CE extended citizenship to virtually all free inhabitants, transforming citizenship from a privilege held by perhaps **4-7% of the empire's population** under Augustus to a universal status for free persons [HIGH confidence].

Slave population estimates: **10-20% of total population**, with Roman Italy containing **1-1.5 million enslaved persons** representing **13-28% of the Italian population** [MODERATE confidence].

C. Gini Coefficient Estimates

Metric	Value	Period	Source	Confidence
Gini coefficient	0.39	14 CE	Milanovic et al. 2007	MODERATE
Gini coefficient	0.41-0.44	150 CE	Scheidel & Friesen 2009	MODERATE
Top 1% income share	16.1%	14 CE	Milanovic et al. 2007	MODERATE
Extraction ratio	75-94%	Imperial period	Milanovic et al. 2007	MODERATE

D. Social Mobility Pathways

Roman manumission practices were notably frequent. Freedmen faced legal bars from holding office themselves but their children possessed full legal rights, enabling second-generation advancement to senatorial status. The *cursus honorum* established formalized career progression with approximately **20 quaestorships** filled annually [HIGH confidence].

II. BYZANTINE EMPIRE (330-1453 CE)

A. Prosopographical Database Coverage

- **PBW (Prosopography of the Byzantine World, 1025-1180):** ~10,000 individuals with ~60,000 factoids
- **PBE:** 641-867 CE
- **PmbZ:** 641-1025 CE (5 volumes)
- **Prosopographisches Lexikon der Palaiologenzeit:** 1261-1453

The Kletorologion of Philotheos (899 CE) documents **60 higher ranks** in court hierarchy, with **50 of these 60 positions available to eunuchs** [HIGH confidence].

B. Eunuch Price Differentials (Social Mobility Indicator)

Category	Price (nomismata)	Confidence
10-year-old child	10	MODERATE
10-year-old castrated	30	MODERATE
Adult male	20	MODERATE
Adult eunuch	50	MODERATE

Premium for eunuchs: **150%** — reflecting anticipated career value [MODERATE confidence].

C. Population Estimates

Period	Population	Confidence
Justinian's peak (541)	25-30 million	MODERATE
Post-7th century losses	~10-12 million	MODERATE
10th century recovery	~12-18 million	MODERATE
After 1204	2-4 million	MODERATE
1453	~500,000 or less	MODERATE

Constantinople: **300,000-500,000** at peak → **~70,000** during Dark Ages → **~50,000** at 1453 fall.

D. Literacy Estimates

Average: ~30% (Brownworth estimate), though "plunged well below 30%" during 7th-8th century crisis [LOW confidence].

E. Theme System

- Themes expanded: 4 (640s-660s) → 29 (10th c.) → 38 (11th c.)
- Total thematic soldiers: ~120,000 circa 775; 100,000-120,000 around 900 CE [MODERATE confidence]

F. Gini Estimate

Byzantine 1000 CE: Gini 0.40-0.45 with highest top 1% income share (30.6%) in Milanovic's pre-modern sample [MODERATE confidence].

III. TANG DYNASTY CHINA (618-907 CE)

A. Aristocratic Clan Structure

Nicolas Tackett's database: ~30,000 individuals reconstructed into ~4,000 patriline from 3,329+ tomb epitaphs. Analysis reveals 87 core patriline representing 60% of known elite individuals, connected through 384 documented marriages [HIGH confidence].

B. Examination System Metrics (BENCHMARK DATA)

Metric	Value	Confidence
Total jinshi awarded (entire Tang)	6,585 degrees	HIGH
Average jinshi per year	~23	HIGH
Candidates per year (jinshi)	1,000-2,000	HIGH
Jinshi pass rate	1-2%	HIGH
Mingjing pass rate	10-20%	HIGH

Chancellor backgrounds over time:

- Emperor Xuanzong (713-756): ~33% examination background

- Emperor Xianzong (806-821): ~60% examination background
- 9th century overall: ~85% examination background

A 2024 PNAS study confirms aristocratic pedigree's effect **declined throughout Tang** while examination success **increased in importance** [HIGH confidence].

C. Land Allocation (Equal-Field System)

Category	Allocation
Adult male (ding)	100 mu total
— Koufen tian (returnable)	80 mu
— Yongye tian (heritable)	20 mu
Infirm/disabled	40 mu
Widows/concubines	30 mu

System collapsed by ~705; formally replaced by Two-Tax System in **780 CE** [HIGH confidence].

D. Gini Estimate

NO SCHOLARLY GINI ESTIMATE EXISTS FOR TANG CHINA — major data gap.

IV. ABBASID CALIPHATE (750-1258 CE)

A. Ethnic Composition Transformation

Under Umayyads: mawali (non-Arab Muslim converts) faced systematic discrimination. Abbasids abolished formal mawali discrimination, opening administrative positions to Persian bureaucratic class.

B. Bulliet Conversion Curve Estimates

Region	~750 CE	~800 CE	~900 CE	~1000 CE	Confidence
Iran	7%	45%	Majority	~80%	MODERATE
Iraq	—	~40%	—	—	LOW
Egypt	—	—	~50%	—	LOW

C. Turkish Military Elite

Al-Mu'tasim (r. 833-842): assembled **at least 7,000 Turkic slave soldiers** (ghulam) — force large enough to cause Baghdad riots and necessitate moving capital to Samarra in 836 [HIGH confidence].

Mamluk price: "**three to four times that of a top warhorse**" [MODERATE confidence].

D. Population Estimates

Baghdad peak (9th century): **800,000-1,200,000** inhabitants — likely world's largest city until 930s [MODERATE confidence].

Slave population of Iraq (late 9th century): **~300,000**; Basra area: **~15,000 slaves** representing **~25% of labor force** [MODERATE confidence].

E. Gini Estimate

NO SCHOLARLY GINI ESTIMATE EXISTS FOR ABBASID CALIPHATE — major data gap.

V. OTTOMAN EMPIRE (1299-1922 CE)

A. Askeri-Reaya Division

- Reaya** (tax-paying subjects): **>90%** of population
- Askeri** (military-administrative, tax-exempt): **<10%** of population

[HIGH confidence]

B. Devşirme System Quantification (CRITICAL DATA)

Metric	Value	Confidence
Total boys collected (15th-17th c.)	200,000-300,000	MODERATE
Collection frequency	Every 4-5 years	HIGH
Collection ratio	1 boy per 40 households	HIGH
Boys per levy	~3,500	MODERATE
Age range	8-18 (typically 8-14)	HIGH

Metric	Value	Confidence
Grand Viziers from Enderun	64	HIGH

Post-1453: Grand Viziers chosen "nearly exclusively from the kul system" — men of "Byzantine or Balkan background" [HIGH confidence].

C. Timar System

37,818 timar holders recorded in 1525 tax rolls [HIGH confidence].

Classification	Annual Revenue (Akçe)
Timar	Under 20,000
Ziamet	20,000-100,000
Has	Over 100,000

Early timariot ethnic composition (1431 Albania): 16% Christian cavalrymen, 30% Turks, 50% Sultan's slaves [HIGH confidence].

D. Janissary Corps Evolution

Period	Numbers	Confidence
1389 (Kosovo)	2,000	MODERATE
Mehmed II	10,000	MODERATE
1514	16,643	HIGH
Suleiman I	~30,000	MODERATE
1680	54,222	HIGH
1694-95	78,798	HIGH

E. Religious Demographics

- Muslims: ~two-thirds of empire population
- Turks: ~40% — minority in their own empire

- Istanbul 16th c.: ~60% Muslim, ~30% Christian, ~10% Jewish
- Jewish population: **150,000** (world's largest by late 16th c.)
- Slave population (Constantinople 16th-17th c.): ~**20%**

[MODERATE confidence]

F. Guild Structure (Istanbul)

Per Evliya Çelebi (17th c.): **158 artisan associations** with **79,764 esnaf members** operating from **23,214 ateliers and shops** [HIGH confidence].

G. Gini Estimate

Ottoman cities (18th c.): **Gini 0.70+** (wealth-based) [MODERATE confidence — Coşgel & Ergene 2012].

VI. VENETIAN REPUBLIC (697-1797 CE)

A. Libro d'Oro and Patrician Tracking (BEST DOCUMENTED)

Libro d'Oro established **1315**, maintained until 1797 — near-complete elite registry for 482 years.

Category	Count	Timeframe
Case Vecchie ("Old Houses")	24-25 families	Pre-800 CE
Case Nuove ("New Houses")	~150 families	800-1297 CE
Case Nuovissime	30 families	After 1381
Case Fatte Per Soldo (Candia)	~75 families	1645-1669
Case Fatte Per Soldo (Morea)	~50 families	1684-1718

B. Great Council Membership Time Series

Year	Members	Notes
1143	35	Counsellors
1297	~950	After Serrata

Year	Members	Notes
1340	1,212	—
1527	2,746	Maximum
Post-1630 plague	1,660	—
1797	~1,100	Final

[HIGH confidence]

C. Elite Closure Metrics

- Pre-Serrata annual family exit rate: **1.2%**
- Post-Serrata annual family exit rate: **0.14%**
- Reduction: **Order of magnitude**

Between 1297-1381: only **15 families** added in 84 years [HIGH confidence].

D. Three-Tier Structure

Class	Percentage	Functions
Nobili (Patricians)	5-7%	Government
Cittadini (Citizens)	7-8%	Bureaucracy
Popolani (Commoners)	85-90%	Labor

E. Gini Estimates (HIGHEST QUALITY DATA)

Community Type	Average Gini
Cities	~0.90
Agro-towns	~0.85
Rural communities	~0.79

Bergamo specifically: **Gini 0.715-0.753** (16th-17th c.) [HIGH confidence — Alfani research].

Venetian taxation was regressive: poorest 10% taxed **5.4-6% of income** while richest 5% paid only **3.9-4.4%** (1550) [HIGH confidence].

F. Doge Family Concentration

118 doges from only **59 families**. Top 8 families produced **43 doges (36%)** [HIGH confidence].

VII. MONGOL EMPIRE (1206-1368 CE)

A. Documentation Limitations

Principal sources: Rashīd al-Dīn's *Jāmi‘ al-Tawārīkh* (c. 1307), *Secret History of the Mongols*, Yuan Chinese records. Empire's unified period: only **54 years** (1206-1260).

B. Chinggisid Lineage

Only Chinggisids from legitimate line could rule under the "Chinggisid Principle" — persisted until 19th century.

Modern Kazakh Tore clan: **~8,000 verified descendants** through documented genealogies [MODERATE confidence].

C. Keshig (Imperial Guard)

Period	Size	Confidence
1203 (founding)	150	HIGH
Early Genghis	1,000	HIGH
By 1206 quriltai	10,000	HIGH
Kublai Khan	12,000	MODERATE

Keshig served as primary elite formation pathway [HIGH confidence].

D. Yuan Four-Class System

Class	Composition	Est. %	Confidence
Mongols	Ruling ethnicity	~1.5%	MODERATE

Class	Composition	Est. %	Confidence
Semu	Central/West Asians	~1.5%	MODERATE
Hanren	Former Jin subjects	~14%	MODERATE
Nanren	Former Song subjects	~83%	MODERATE

1290 Yuan census: ~10 million Hanren + ~60 million Nanren = ~58 million taxed [MODERATE confidence].

E. Examination Restoration (1315)

Equal quotas: 75 provincial + 25 metropolitan per class = 300 total jinshi, though difficulty intentionally lower for Mongols and Semu [HIGH confidence].

F. Fundamental Demographic Reality

Mongol homeland: ~800,000 people. Empire at peak: >100 million inhabitants. Yuan China: Mongols = ~1.5% of population — ratio of ~100:1 ruled to rulers [MODERATE confidence].

G. Gini Estimate

NO SCHOLARLY GINI ESTIMATE EXISTS FOR MONGOL EMPIRE — major data gap.

VIII. CROSS-EMPIRE COMPARISON TABLES

Table 1: Gini Coefficient Estimates

Empire	Period	Gini	Type	Top 1 % Share	Confidence
Roman	14 CE	0.39	Income	16.1%	MODERATE
Roman	150 CE	0.41-0.44	Income	~20%	MODERATE
Byzantine	1000 CE	0.40-0.45	Income	30.6%	MODERATE
Tang	—	NO ESTIMATE	—	—	—
Abbasid	—	NO ESTIMATE	—	—	—
Ottoman	18th c.	0.70+	Wealth	—	MODERATE
Venetian	1400-1800	0.70-0.90	Wealth	—	HIGH

Empire	Period	Gini	Type	Top 1 % Share	Confidence
Mongol	—	NO ESTIMATE	—	—	—

Table 2: Elite Composition (Hereditary vs. Merit)

Empire	Hereditary Component	Merit Component	Confidence
Roman	Senatorial order (1M sesterces)	Freedmen advancement	MODERATE
Byzantine	Dynatoi increasingly hereditary	Eunuch positions (50/60)	MODERATE
Tang	Great clans; yin privilege	Jinshi exam (1-2% pass)	HIGH
Abbasid	Arab lineage (declining)	Mawali; mamluk system	MODERATE
Ottoman	Turkish families (pre-1453)	Devşirme (200K-300K)	HIGH
Venetian	Libro d'Oro (closed 1297)	Aggregazioni only under crisis	HIGH
Mongol	Chinggisid principle	Keshig advancement	MODERATE

Table 3: Social Mobility Indicators

Empire	Primary Mechanism	Quantified Rate	Confidence
Roman	Manumission	"Frequent" (unquantified)	MODERATE
Byzantine	Eunuch positions	50/60 positions	MODERATE
Tang	Jinshi examination	1-2% pass rate	HIGH
Abbasid	Mawali integration	Bulliet curve	MODERATE
Ottoman	Devşirme	1/40 households	HIGH
Venetian	Aggregazioni	0.14%/year family exit	HIGH
Mongol	Keshig service	10,000 members	MODERATE

IX. DATA GAPS SUMMARY

Empires Without Gini Estimates

- Tang Dynasty
- Abbasid Caliphate
- Mongol Empire

Categories Lacking Data (All Empires)

- Literacy rates by class
- Life expectancy by class
- Land ownership Gini coefficients
- Cross-class marriage rates
- Precise manumission/mobility rates

Best-Documented Metrics for Formalization

1. Venice Libro d'Oro (near-complete family tracking)
2. Tang examination statistics (pass rates, degree totals)
3. Ottoman devşirme records (collection ratios)
4. Roman property qualifications (senatorial/equestrian thresholds)
5. Alfani's Venetian wealth Ginis (archival estimi analysis)

X. SCHOLARLY SOURCES

Cross-Empire: Branko Milanovic, Walter Scheidel, Guido Alfani, Peter Turchin

Roman: Keith Hopkins (*Death and Renewal*), PLRE database, Peter Garnsey

Byzantine: Alexander Kazhdan, PBW/PBE databases, Warren Treadgold

Tang: Nicolas Tackett (*Destruction of Medieval Chinese Aristocracy*), PNAS 2024 study

Abbasid: Richard Bulliet (conversion curves), Roy Mottahedeh

Ottoman: Metin Kunt, Halil İnalcık, Evliya Çelebi, Coşgel & Ergene

Venetian: Stanley Chojnacki, Guido Alfani, Dorit Raines

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